

ADITYA KULKARNI

(281) 876-7066

✉ adityakulkarnius@gmail.com [in linkedin.com/in/aditya-kulkarni-355b81217](https://www.linkedin.com/in/aditya-kulkarni-355b81217) github.com/Aditya-k24 kulkarniaditya.com

EDUCATION

North Carolina State University

M.S. Computer Science

Raleigh, NC

Aug 2024 – May 2026

University of Mumbai

B.Tech. Computer Science, CGPA: 9.32/10

Mumbai, India

Aug 2020 – May 2024

WORK EXPERIENCE

Product Engineer Intern

May 2025 – Present

Isomer AI

Remote

- Shipped 10+ production backend features for an AI-powered financial automation platform – Temporal.io workflow orchestration, RBAC, and real-time event pipelines – across 7 deployments using TypeScript, Node.js, Express, PostgreSQL, and Redis
- Integrated document processing pipelines (PDF generation, email parsing via Microsoft Graph API, S3 storage) and multi-model LLM tooling across 3 providers (Claude, GPT-4o, Gemini); contributed 394 commits and led code review in high-velocity Agile sprint cycles

Graduate Research Assistant

Apr 2025 – Present

NC State University

Raleigh, NC

- Engineered a cloud-based genomic data platform on Google Cloud Run with JBrowse 2, enabling interactive visualization of 500GB datasets for 100+ researchers worldwide
- Improved ML model accuracy 12% through hyperparameter tuning and time-series analysis on biological datasets

Software Developer Intern

Jun 2024 – Aug 2024

Mphasis (HP Company)

Remote

- Built 5+ AI-driven automation workflows including a health claim processing tool using OCR, NLP, and ML models, improving claim accuracy by 25% in production
- Automated CI/CD deployments on AWS and GCP, integrated Grafana monitoring dashboards, and supported model validation to enhance production reliability

PERSONAL PROJECTS

KubeServe – Kubernetes LLM Inference Platform | Python, FastAPI, Redis, Kubernetes, KEDA, Prometheus 2026

- Built a Kubernetes-native LLM inference platform: FastAPI router with 3-state circuit breaker and EWMA latency-aware load balancing across Claude, GPT-4o, and Gemini backends; KEDA autoscaler scales model pods 0→N on Redis queue depth
- Implemented Python Kopf CRD operator reconciling Deployments, Services, and KEDA ScaledObjects; OpenTelemetry + Prometheus + Grafana observability; k6 load-tested to 500 VUs; 88 pytest tests; Helm + ArgoCD GitOps

PulseRank – Real-Time Leaderboard & Activity Feed | TypeScript, NestJS, MySQL, MongoDB, Redis 2026

- Built event-driven leaderboard and activity feed: MySQL for transactional writes with idempotency-key deduplication and outbox pattern, MongoDB for feed projections with TTL expiry, Redis sorted sets for sub-millisecond leaderboard reads via Server-Sent Events
- Validated multi-store consistency with Testcontainers integration tests, benchmarked hot paths with k6, shipped CI/CD with GitHub Actions, and instrumented OpenTelemetry + Prometheus for full observability

Inventory Reservation Service | Java 21, Spring Boot, PostgreSQL, Testcontainers, Docker 2026

- Built a transactional inventory reservation REST API in Java 21 and Spring Boot preventing overselling via PostgreSQL row-level locking (SELECT FOR UPDATE) across multi-SKU reservations with idempotent endpoints and RFC 9457 error responses
- Verified concurrency correctness with Testcontainers – two concurrent requests against quantity 1 produce exactly one success and one 409 Conflict; Flyway migrations, Spring Security RBAC, multi-stage Docker build, GitHub Actions CI

TECHNICAL SKILLS

Languages: TypeScript, Java, Python, JavaScript, SQL, NoSQL, C, Ruby

Backend & APIs: Node.js, Express, NestJS, Spring Boot, FastAPI, REST APIs, GraphQL

Databases: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch

Infrastructure: AWS, GCP, Docker, Kubernetes, GitHub Actions, CI/CD, Terraform, Grafana

Frameworks & Tools: Temporal.io, Kafka, KEDA, LangChain, PyTorch, TensorFlow